carries the two sperm to the micropyle (small opening) of an ovule. Flowering plants undergo **double fertilization:** One sperm unites with an egg, forming a diploid zygote, and the other unites with polar nuclei, forming a triploid endosperm nucleus (see Section 27.1).

Ultimately, the ovule becomes a seed that contains the embryo (the sporophyte of the next generation) and stored food enclosed within a seed coat (see Fig. 23.2*b*). Endosperm in some seeds is absorbed by the cotyledons, whereas, in other seeds, endosperm is digested as the seed matures.

A **fruit** is derived from an ovary, and in some instances it is an accessory part of the flower. Some fruits, such as apples and tomatoes, provide a moist, fleshy covering; other fruits, such as pea pods and acorns, provide a dry covering for seeds.

Flowers and Diversification

Flowers are involved in the production and development of spores, gametophytes, gametes, and embryos enclosed within seeds. Successful completion of sexual reproduction in angiosperms requires the effective dispersal of pollen and then seeds. The various ways pollen and seeds can be dispersed have resulted in many different types of flowers (see the Evolution feature, "Plants and Their Pollinators," in Section 27.1).

Wind-pollinated flowers are usually not showy, whereas many insect- and bird-pollinated flowers are colorful. Night-blooming flowers attract nocturnal mammals or insects; these flowers are usually aromatic and white or cream-colored.

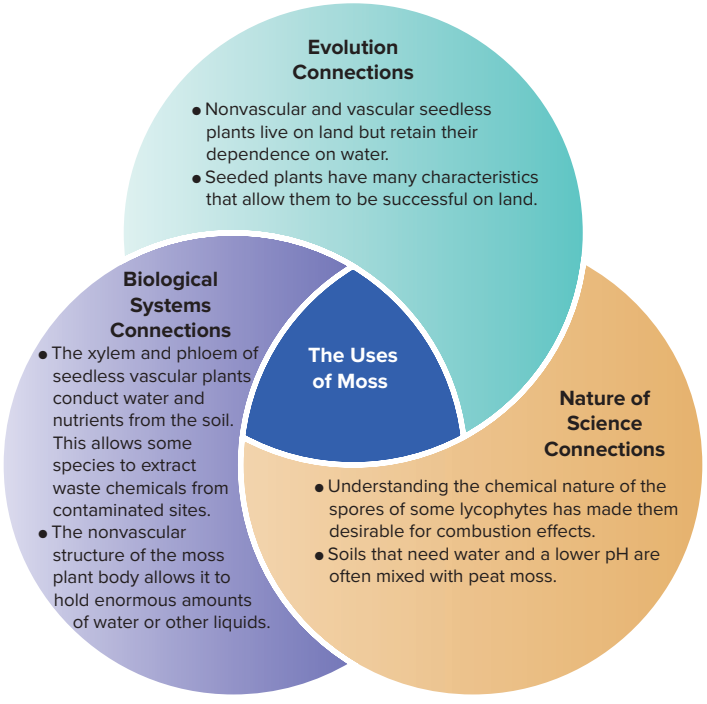
Although some flowers disperse their pollen by wind, many are adapted to attract specific pollinators, such as bees, wasps, flies, butterflies, moths, and even bats, that carry pollen from one flower to another flower of the same type. For example, glands in the region of the ovary produce nectar, a nutrient that is gathered by pollinators as they go from flower to flower. Bee-pollinated flowers are usually blue or yellow and have ultraviolet shadings that lead the pollinator to the location of the nectar. The mouthparts of bees are fused into a long tube, which is able to obtain nectar from the base of the flower. In another example, instead of beautiful colors and tempting nectar, some species of *Ophyrys* orchids use "sexual mimicry" to attract their wasp pollinators. The orchid flowers look like a female wasp and emit pheromones to attract male wasps. The males will engage in "pseudocopulation" with the "female" and end up with pollen attached to its head. Frustrated, the male wasp leaves the flower and attempts to mate with another flower. The male then deposits pollen to the second flower, thereby completing the cross-pollination that the orchid intended.

The fruits of flowers protect and aid in the dispersal of seeds. Dispersal occurs when seeds are transported by wind, gravity, water, and animals to another location. Fleshy fruits may be eaten by animals, which transport the seeds to a new location and then deposit them when they defecate. Because animals live in particular habitats and/or have particular migration patterns, they are apt to deliver the fruit-enclosed seeds to a suitable location for seed germination (when the embryo begins to grow again) and development of the plant.

Check Your Progress 23.5

1. List the life cycle changes that have enabled pines to better adapt to life on land.
2. Compare and contrast the four types of gymnosperms.
3. List the functions of the key structures required for angiosperm reproduction.

Connecting *the* Concepts



ISTUDY